

From Pixels to Hierarchical Sequences: Quadtree Mask Encoding for Vision-Language Binary Change Detection

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Abstract

Dense change detection in remote sensing requires vision-language models (VLMs) to compare bi-temporal images and generate accurate pixel-level masks. Existing VLMs are largely confined to change captioning outputs, and the few that produce pixel-level masks still rely on external decoders or flat text-as-mask serialization, which are less effective for small and fragmented changes. We introduce QUAKE-CD, a framework that recasts dense change prediction as syntax-verifiable structured generation. QUAKE-CD represents binary change masks as grammar-constrained quadtree token sequences, making the masks compact, syntactically checkable, and deterministically decodable within an autoregressive generation space. We further construct QUAKE-CoT, which pairs these sequences with chain-of-thought traces grounded in visual evidence, and jointly optimizes textual reasoning and spatial dense prediction through a progressive curriculum followed by grammar-gated dual-reward RL. On QUAKE-CoT, QUAKE-CD achieves 78.31% accumulated F1, outperforming decoder-based and flat text-as-mask VLMs while producing more faithful bi-temporal reasoning.

1 Introduction

Large vision-language models (VLMs) demonstrate profound capability in multimodal reasoning (Bai et al. 2025; Team et al. 2026), yet real-world applications increasingly demand spatially grounded outputs that delineate precise footprint boundaries (Xiao et al. 2025; Ding et al. 2025). Remote sensing change detection (Asokan and Anitha 2019; Dong et al. 2024) provides a stringent testbed for this imperative. Given bi-temporal images, models must synthesize cross-temporal visual evidence to render a pixel-level binary change map. Traditional change detection paradigms, including encoder-decoder networks and open-vocabulary formulations, narrowly optimize for discriminative mask prediction (Dong et al. 2024; Korkmaz and Patel 2026; Zhu et al. 2025), structurally divorcing the semantic interpretation of changes from spatial footprints. While recent VLMs support interactive change captioning and localization (Deng, Zhou, and Wu 2025; Li et al. 2026b; Elgendy et al. 2024), *natively* generating complete, dense spatial masks remains challenging. This limitation exposes a critical dichotomy between

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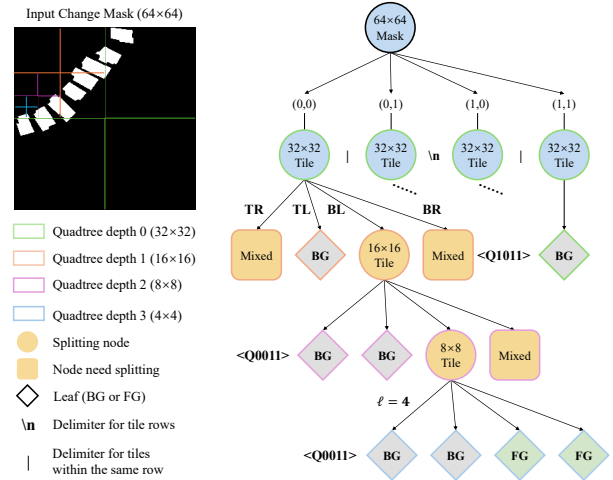


Figure 1: Illustration of the proposed QUAKE grammar. We take 64×64 change mask and $\ell = 4$ as an example.

language-centric reasoning and pixel-level decision-making.

Rather than bridging this gap, contemporary VLM architectures often relegate dense prediction to an auxiliary objective, mapping hidden states to spatial masks via task-specific decoders (Lai et al. 2024; Ren et al. 2024; Ma et al. 2025; Li et al. 2026a), which remains fundamentally orthogonal to the core autoregressive mechanism. While effective in isolation, this decoupling necessitates heterogeneous objectives, encumbering end-to-end training and stifling direct token-level optimization. Furthermore, sparse-cue methods that delegate mask generation to external segmenters (e.g., SAM (Kirillov et al. 2023)) via bounding boxes or points (Sun et al. 2026; Yao et al. 2026; Liu et al. 2025) not only lack native support for bi-temporal contexts but also falter on multiple fragmented instances. These coarse prompts inevitably entangle subtle targets with irrelevant background, triggering cascading localization errors that amplify false predictions.

Integrating dense prediction natively into VLMs requires formulating output masks as structured, autoregressive language. The emerging *text-as-mask* paradigm (Lan et al. 2025, 2026; Liu and Lian 2026) offers a promising trajectory by serializing spatial layouts into textual descriptors. However, existing methods relying on flat patch-grid serialization treat

all spatial regions equally, misaligning with sparse change masks where fine-grained boundaries demand high token density. Furthermore, this uniform encoding causes densely distributed targets in remote sensing (RS) imagery to suffer from severe spatial adhesion, erroneously collapsing distinct instances into a single connected component. Consequently, the task necessitates a hierarchical representation capable of efficiently compressing uniform regions while adaptively allocating tokens to areas demanding high spatial acuity.

We introduce QUAKE-CD, a framework that unifies language-centric reasoning with pixel-level prediction by encoding binary masks via **Quadtree mask Encoding (QUAKE)**—a hierarchical spatial language shown in Figure 1 that supports dense spatial optimization directly within the native autoregressive token space. By recursively refining heterogeneous regions and collapsing uniform areas, *QUAKE* adaptively allocates tokens to preserve fine-grained boundaries while compressing redundant background. Crucially, its grammar makes structural errors observable, transforming dense prediction into a syntax-verifiable language generation that aligns naturally with explicit reasoning traces.

Trained exclusively on *QUAKE*, VLMs mechanically memorize the syntax but struggle to grasp the underlying semantics, degenerating into hollow pattern matching. To explicitly guide the model and anchor this grammar in genuine visual understanding, we construct *QUAKE-CoT*, an instruction dataset that couples *QUAKE* sequences with chain-of-thought (CoT) rationales detailing the type, location, and extent of changes. To avert optimization conflicts, our progressive curriculum decouples these dual objectives: the model first masters valid sequence generation before integrating the bi-temporal rationales. Exploiting the syntactic verifiability of *QUAKE*’s textual form, we further align the model via grammar-gated dual-reward reinforcement learning (RL). A rule-based *Quadtree Reward* validates syntax and scores decoded masks against ground truth using a Tversky-style spatial objective, preventing format collapse while optimizing dense prediction quality. In parallel, *ThinkAnswer Reward* evaluates whether the reasoning trace faithfully reflects the observed changes. Optimizing these rewards with GRPO (Guo et al. 2025) jointly enforces grammaticality, pixel-level accuracy, and visually grounded reasoning.

Experiments show that QUAKE-CD substantially improves pixel-level VLM change detection. It achieves 78.31% accumulated F1 and 73.18% per-image F1, surpassing decoder-based baselines and uniform text-as-mask serializations. The results expose two failure modes of existing designs: decoder-based VLMs produce spatial outputs weakly coupled to cross-temporal evidence, while uniform grid serialization over-covers small, densely distributed changes. Reasoning evaluations further confirm that grounded spatial supervision yields more faithful bi-temporal explanations.

In summary, our main contributions are:

- We introduce *QUAKE*, a hierarchical text-as-mask language whose context-free grammar admits linear-time syntactic and structural validation, embedding dense spatial prediction into the autoregressive token space.
- We construct *QUAKE-CoT*, a dataset that explicitly cou-

ples hierarchical mask sequences with CoT rationales, and establish a progressive curriculum to anchor spatial grammar in genuine bi-temporal understanding.

- We develop a grammar-gated dual-reward RL paradigm that jointly optimizes format validity, pixel-level mask accuracy, and semantic reasoning fidelity within a unified token-level optimization process.

2 Related Work

Pixel-Level Prediction in RS Vision-Language Models.

While VLMs have been widely adapted for RS applications including change captioning (Deng, Zhou, and Wu 2025), change grounding (Xue et al. 2026), and RL-driven spatial reasoning (Yao et al. 2026; Sun et al. 2026), achieving precise pixel-level interaction remains a systemic challenge. Most approaches either employ auxiliary mask decoders (Lai et al. 2024; Ren et al. 2024; Li et al. 2025) or generate sparse coordinate cues that inevitably delegate dense mask prediction to external segmenters (Sun et al. 2026; Liu et al. 2025). Although the emerging *text-as-mask* paradigm bypasses external modules by serializing spatial layouts directly into text (Lan et al. 2025; Liu and Lian 2026), its rigid, flat-grid formulation fails to capture fine boundaries and consistently induces severe spatial adhesion among dense semantic targets. QUAKE-CD fundamentally diverges by formulating the bi-temporal change mask as a grammar-constrained hierarchical *QUAKE* sequence, adaptively concentrating tokens along boundaries to ensure native, precise mask generation without external visual dependencies.

Grammar-Aligned Spatial Reasoning. Standard CoT prompting inherently struggles with complex spatial comprehension (Shiri et al. 2024), routinely hallucinating rationales without explicit grounding supervision (Chen et al. 2024). While recent advances employ grammar constraints for structural validity (Park et al. 2024) and RL to refine reasoning trajectories (Guo et al. 2025), these mechanisms operate in isolation: reasoning is evaluated as unstructured text, grammar superficially restricts decoding, and RL predominantly targets sparse bounding-box objectives (Sun et al. 2026; Yao et al. 2026). QUAKE-CD departs from this line by inextricably anchoring the semantic rationale to a formal *QUAKE* grammar; the resulting dual-reward RL framework jointly enforces grammatical determinism, semantic faithfulness, and pixel-level precision.

3 Methodology

Problem Formulation

QUAKE-CD reformulates bi-temporal change detection as autoregressive structured generation. Given a bi-temporal image pair (I_1, I_2) , a VLM π_θ generates a tripartite token sequence $\mathbf{y} = [\mathbf{y}_{\text{think}}; \mathbf{y}_{\text{answer}}; \mathbf{y}_{\text{seg}}]$, where $\mathbf{y}_{\text{think}}$ constitutes a free-text CoT reasoning trace, $\mathbf{y}_{\text{answer}}$ provides a concise change summary, and $\mathbf{y}_{\text{seg}}$ acts as a quadtree encoding drawn from a formal language $\mathcal{L}_Q \subset V^*$. A deterministic decoder $\mathcal{D} : \mathcal{L}_Q \rightarrow \{0, 1\}^{H \times W}$ then maps any valid encoding back to a binary change mask $\hat{M} = \mathcal{D}(\mathbf{y}_{\text{seg}})$.

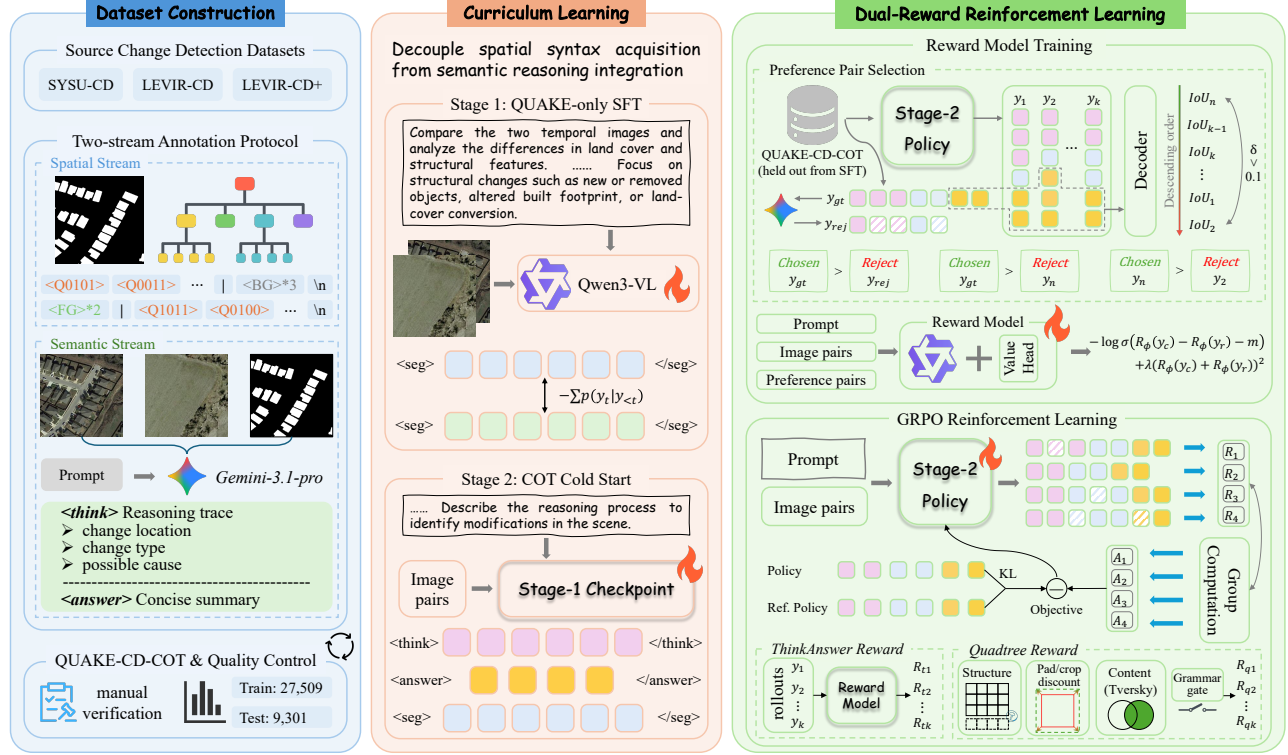


Figure 2: Illustration of the QUAKE-CD Pipeline. (1) Construction of the QUAKE-CoT dataset via a two-stream annotation protocol. (2) Two-stage progressive SFT curriculum. (3) Reward model training and grammar-gated dual-reward RL.

Quadtree Mask Encoding

Vocabulary and Encoding Grammar. The language $\mathcal{L}_Q$ is defined over an alphabet of special tokens integrated into the VLM vocabulary. This alphabet encompasses wrapper delimiters $\langle \text{seg} \rangle$ and $\langle / \text{seg} \rangle$, uniform tile tokens $\langle \text{BG} \rangle$ representing pure background and $\langle \text{FG} \rangle$ denoting pure foreground, alongside fifteen quadtree node tokens $\langle \text{Q0001} \rangle$ through $\langle \text{Q1111} \rangle$. The all-zero pattern $\langle \text{Q0000} \rangle$ is explicitly excluded as it inherently signifies an empty quadrant.

The syntax of $\mathcal{L}_Q$ obeys a context-free grammar $\mathcal{G}_Q$. A valid encoding conforms to the structure $\langle \text{seg} \rangle G \langle / \text{seg} \rangle$, where G specifies a grid of non-overlapping 32×32 image tiles ordered via raster scan. Tiles within a row are delineated by $|$ symbols, while distinct rows are separated by newline characters. This 32×32 tile dimension aligns perfectly with the native visual token granularity of Qwen3-VL (Bai et al. 2025), ensuring the output grid shares an identical spatial architecture with the visual feature map. Within each row, consecutive identical uniform tiles undergo run-length compression as $\langle \text{BG} \rangle * n$ or $\langle \text{FG} \rangle * n$ for integer $n \geq 2$.

As outlined in Figure 1 and Algorithm 1, given a binary change mask $M \in \{0, 1\}^{H \times W}$, a deterministic encoder $E : \{0, 1\}^{H \times W} \rightarrow \mathcal{L}_Q$ partitions M into the aforementioned tile grid. Each tile acts either as a uniform token or a quadtree expansion represented by a node $\langle \text{Q}abcd \rangle$. The four constituent bits specify which respective quadrants contain foreground pixels across the top-left, top-right, bottom-left, and bottom-right sectors. Active quadrants undergo recursive

preorder expansion until reaching a minimum leaf size ℓ .

The grammar $\mathcal{G}_Q$ is context-free with bounded recursion depth $d_{\max} = \lceil \log_2(32/\ell) \rceil$. We further distinguish three levels of correctness: **syntactic validity**, certified in linear time by a single-pass parser under $\mathcal{G}_Q$; **structural validity**, verified during decoding by matching the reconstructed grid against the expected $H/32 \times W/32$ layout, and **semantic correctness**, optimized against the ground truth through the reward framework in Section 3. Crucially, this encoding is **recall-preserving** by construction: any leaf cell containing a foreground pixel dictates a foreground encoding, strictly guaranteeing $\forall(i, j)$ where $M[i, j] = 1$, $\mathcal{D}(E(M))[i, j] = 1$. The complementary precision loss is strictly bounded by ℓ , restricting over-encoding solely to boundary cells mixing foreground and background. The parameter ℓ offers explicit modulation of this precision-compression trade-off, with concrete value selections analyzed in Section 4.

Mask Decoding. The decoding function $\mathcal{D} : \mathcal{L}_Q \rightarrow \{0, 1\}^{H \times W}$ effectively reverses E by parsing the *QUAKE* sequence, resolving run-length compression, and recursively rendering quadtree nodes back into 32×32 pixel blocks. Consequently, $\mathcal{D}$ operates as a purely **non-parametric** mapping, circumventing neural networks and ensuring high efficiency during both RL reward computation and evaluation.

Supervised Fine-Tuning with Curriculum

Dataset Construction. Existing change detection benchmarks, SYSU-CD (Shi et al. 2021), LEVIR-CD and LEVIR-

CD+ (Chen and Shi 2020), provide pixel-level masks but no language annotations. We assemble QUAKE-CoT from these sources via a two-stream annotation protocol. As illustrated in Figure 2, the **spatial stream** deterministically encodes each ground-truth mask as $E(M) \in \mathcal{L}_Q$. The **semantic stream** prompts *Gemini-3.1-Pro-Thinking-Preview* (Team et al. 2023), conditioned on the bi-temporal pair and ground-truth mask, to produce a reasoning trace covering change type, location, extent, and plausible cause, followed by a concise summary. Dataset details and system prompts are shown in Appendix B and G.

These independently generated streams are subsequently paired into samples of the form `<think> reasoning </think> <answer> summary </answer> <seg> QUAKE </seg>`. Mask-conditioned generation makes these traces a **grounding supervision signal** that anchors *QUAKE* syntax in visually verifiable semantics, not a record of naturalistic inference. The original training splits yield 27,509 training samples; the union of validation and test splits forms a 9,301-sample test set, precluding leakage. Quantitative assessments regarding the round-trip fidelity of E and $\mathcal{D}$ on the QUAKE-CoT dataset are established in Table 3. To our knowledge, QUAKE-CoT constitutes the first instruction-tuning dataset for bi-temporal change detection that pairs structured reasoning traces with formalized, deterministically decodable dense mask encodings.

Curriculum Stage-1: QUAKE-Only SFT. Attempting simultaneous acquisition of *QUAKE* grammar alongside free-text reasoning precipitates destructive interference, forcing the model to fracture capacity between syntax mastery and semantic analysis. *Stage-1* intentionally isolates spatial syntax acquisition. We fine-tune the model to exclusively generate the `<seg>` block conditioned on a bi-temporal image pair, dedicating its entire generation budget to mastering grid dimensions, uniform tile semantics, recursive node expansion, run-length encoding, and row delimiters.

Curriculum Stage-2: CoT Cold-Start. *Stage-2* integrates reasoning by fine-tuning the Stage-1 checkpoint to synthesize the complete tripartite output. This *cold-start* strategy successfully preserves the entrenched *QUAKE* syntax while catalyzing the acquisition of cross-temporal reasoning capabilities. Subsequent ablations in Section 4 validate the efficacy of this curriculum. The sequential dependency linking these stages operationalizes the curriculum’s central principle: demanding mastery of the spatial alphabet before permitting the composition of spatially grounded visual explanations.

Grammar-Gated Dual-Reward RL

While SFT yields syntactically valid *QUAKE* sequences paired with grounded reasoning, it fundamentally lacks the mechanism to directly optimize the composite objective encompassing format validity, pixel-level accuracy, and reasoning faithfulness. We resolve this limitation with a dual-reward RL framework to jointly optimize dense spatial outputs and strengthen the reasoning trace as a semantic guide for *QUAKE* generation within the autoregressive space.

Reward Model Construction. To train a dedicated reward model for evaluating semantic reasoning faithfulness, we construct preference pairs from the QUAKE-CoT split held

out from SFT. The cold-start model generates multiple completions per sample under high-temperature decoding. Each completion’s corresponding *QUAKE* sequence is decoded via $\mathcal{D}$ and scored against the ground-truth mask M using Intersection over Union (IoU). In parallel, we synthesize one hard-negative response y_{neg} per sample by prompting *Gemini-3.1-Pro-Thinking-Preview* to perturb core semantic facts while preserving fluency and the grammatical structure.

Let $s(y) = \text{IoU}(\mathcal{D}(y_{\text{seg}}), M)$ denote the decoded IoU of a rollout. Given a candidate pool $\mathcal{C} = \{y_{\text{gt}}\} \cup \{y_k\}_{k=1}^K \cup \{y_{\text{neg}}\}$, where y_{gt} is the ground-truth reasoning trace, and $\{y_k\}_{k=1}^K$ denotes K empirical rollouts, we form preference pairs following the five-tier hierarchy in Appendix C, with y_{gt} and y_{neg} serving as fixed top and bottom anchors, and retain rollout pairs whose IoU gap exceeds the threshold δ :

$$\mathcal{D}_{\text{pref}} = \{(y_c, y_r) \in \mathcal{C} \times \mathcal{C} \mid y_c \succ y_r\}, \quad (1)$$

The reward model R_ϕ is trained on $\mathcal{D}_{\text{pref}}$ with Qwen3-VL-8B-Instruct under the objective:

$$\mathcal{L}_{\text{RM}} = -\mathbb{E}_{\mathcal{D}_{\text{pref}}} [\log \sigma(R_\phi(y_c) - R_\phi(y_r)) - \lambda(R_\phi(y_c) + R_\phi(y_r))^2], \quad (2)$$

where y_c and y_r indicate chosen and rejected responses, respectively, and R_ϕ evaluates the semantic caliber of the `<think><answer>` segments. Details on negative generation and RM training dynamics are provided in **Appendix C. ThinkAnswer Reward.** To systematically enforce the semantic validity of the underlying logic during RL, the *ThinkAnswer Reward* R_t acts as a direct application of the previously trained reward model R_ϕ . It specifically scores the `<think>` reasoning trace and `<answer>` summary segments. This incorporates query-wise normalization to guarantee stable optimization signals across highly diverse prompts. Operating as a crucial semantic regularizer, this reward signal discourages structurally valid but factually unfaithful explanations from receiving inflated high scores.

Quadtree Reward. The *Quadtree Reward* R_q operates as a rigorous rule-based function evaluating spatial mask quality natively from the generated sequence. A *grammar gate* validates the `<seg>` sequence against $\mathcal{G}_Q$ via a single-pass linear-time parser. Concurrently, a *structure score* mandates grid dimension alignment with the ground truth. The subsequent *content score* calculates the Tversky index (Tversky 1977) between the decoded prediction $\hat{M}$ and ground truth M . Additionally, a *pad/crop discount* penalizes topological dimension mismatches necessitating fallback alignment, while an *empty-GT guard* outright neutralizes the trivial exploitation strategy of perpetually predicting pure background (see details in **Appendix C**).

GRPO Optimization. We optimize the VLM policy starting from the cold-start checkpoint utilizing the GRPO objective (Guo et al. 2025), using the reward:

$$R(\mathbf{y}, M) = (1 - \mu)R_q(\mathbf{y}, M) + \mu R_t(I_1, I_2, \mathbf{y}) \quad (3)$$

under KL regularization against the reference policy π_{ref} , where μ dictates the weight of the semantic reward. This dual-reward architecture instantiates the central hypothesis of this work: formulating dense spatial outputs as structured language enables joint, token-level optimization of format validity, pixel-level accuracy, and reasoning faithfulness.

Method	Accumulated Mean			Per-Image Mean			Throughput (pairs/s)
	Precision	Recall	F1	Precision	Recall	F1	
BIT	77.08	76.80	76.94	66.34	66.64	63.19	50.25
ChangeFormer	84.23	78.84	81.44	76.61	71.06	70.88	65.78
ChangeCLIP	84.32	78.50	81.30	57.31	50.58	51.04	74.93
LISA-7B	16.65	73.42	27.14	18.04	46.83	21.91	40.21
LISA-llama2-13B	15.04	76.45	25.13	17.10	49.20	21.45	27.15
GLaMM	14.85	52.07	23.11	16.22	29.59	17.44	18.77
PixelLM	16.79	72.51	27.27	17.39	46.38	21.38	14.01
Text4Seg	14.52	84.07	24.76	15.27	54.88	19.65	86.54
RSUniVLM	50.12	82.45	62.34	48.76	81.23	60.94	82.30
QUAKE-CD	70.99	87.31	78.31	68.77	84.09	73.18	74.27

Table 1: Dense change detection performance on the QUAKE-CoT test set. We compare against specialist change detectors (BIT, ChangeFormer, and ChangeCLIP), decoder-based pixel-level VLMs (LISA, GLaMM, PixelLM), a flat text-as-mask method (Text4Seg), and a remote sensing VLM (RSUniVLM) with native bi-temporal input and text-as-mask serialization.

4 Experiments

Experimental Setup

Setup. All experiments utilize the QUAKE-CoT established in Section 3. We allocate 80% of the training set to the two-stage curriculum and reserve the remaining 20% exclusively for reward model training and RL. The framework is initialized from Qwen3-VL-8B-Instruct (Bai et al. 2025). Training is executed via ms-swift (Zhao et al. 2024) on eight NVIDIA H200 GPUs, while inference is served through vLLM (Kwon et al. 2023). Both SFT stages and the reward model employ LoRA (Hu et al. 2022) with rank 64, with the reward model initialized from a fresh backbone copy. The RL phase optimizes the policy via GRPO (Guo et al. 2025) using eight rollouts per prompt, and a KL coefficient of 0.04. Appendix D catalogs the comprehensive hyperparameter specifications.

Baselines. We compare against three complementary baseline groups. First, we include representative specialist remote sensing change detectors: BIT (Chen, Qi, and Shi 2021), ChangeFormer (Bandara and Patel 2022), and ChangeCLIP (Dong et al. 2024). Second, we evaluate pixel-level VLMs across two prevailing paradigms. LISA (Lai et al. 2024), GLaMM (Rasheed et al. 2024), and PixelLM (Ren et al. 2024) perform decoder-based mask prediction, while Text4Seg (Lan et al. 2025) and RSUniVLM (Liu and Lian 2026) serialize masks autoregressively. Specialist change detectors and RSUniVLM are retrained on the QUAKE-CoT training split. RSUniVLM natively accepts RS bi-temporal pairs, while the remaining pixel-level VLMs assume single-image inputs. Following (Nguyen et al. 2025; Xu, Zhu, and Yang 2025), we concatenate the bi-temporal pair (I_1, I_2) along the width dimension and fine-tune them on the QUAKE-CoT training split under their original training recipes. Third, for reasoning-quality evaluation, we include TeoChat (Irvin et al. 2025) and EarthDial (Soni et al. 2025). These temporal RS VLMs generate textual change descriptions but do not predict dense masks. **Additional experimental results on large images, out-of-domain dataset, qualitative interpretability, and parameter sensitivity are reported in Appendix E.**

Dense Change Mask Generation

For each prediction, we decode the extracted *QUAKE* span into a binary mask with $\mathcal{D}$ and report Precision, Recall, and F1 under accumulated pixel statistics and per-image means. As shown in Table 1, QUAKE-CD attains specialist-level mask quality while retaining a unified autoregressive interface for dense prediction and grounded reasoning. Its accumulated F1 of 78.31% surpasses BIT by 1.37% and remains within 3.13% of the strongest specialist model, ChangeFormer. Under per-image evaluation, QUAKE-CD reaches 73.18% F1 and outperforms all three specialist models. It further achieves the highest recall among all methods under both evaluation protocols. Against pixel-level VLMs, QUAKE-CD surpasses PixelLM by 51.04% in accumulated F1 and LISA-7B by 51.27% in per-image F1. Existing VLM baselines exhibit a pronounced recall-precision imbalance: they recover broad change regions but introduce extensive false positives. Even the strongest VLM baseline reaches only 50.12% accumulated precision and 48.76% per-image precision. Text4Seg exemplifies this regime, attaining 84.07% accumulated recall but only 14.52% precision and thus 24.76% F1: its flat patch-grid serialization assigns uniform resolution across the scene and cannot resolve the small, fragmented objects characteristic of change detection. In contrast, QUAKE-CD raises precision to 70.99% and 68.77% while preserving strong recall at 87.31% and 84.09%. This advantage stems from the selective allocation of quadtree tokens around change boundaries, which suppresses spurious regions without erasing fine-grained structures. RSUniVLM adopts flat text-as-mask formulation while accepting the bi-temporal pair natively. Its persistent recall-precision imbalance isolates the QUAKE-CD gain to hierarchical quadtree encoding rather than to native bi-temporal input.

QUAKE-CD also achieves a favorable balance between accuracy and efficiency. Its throughput reaches 74.27 pairs/s, at least $1.85\times$ that of decoder-based VLMs, as native text generation avoids the overhead of a segmentation decoder. Its throughput remains close to flat serialization methods, including Text4Seg at 86.54 pairs/s and RSUniVLM at 82.30 pairs/s, despite the additional sequential structure introduced

Model	BLEU-1	ROUGE-1	METEOR	Reranker	GPT-Judge	Human
Qwen3-VL-8B-Instruct	3.82	23.06	11.08	27.76	47.78	42.25
Qwen3-VL-32B-Instruct	11.90	29.78	15.42	32.41	52.91	51.10
Qwen3-VL-235B-Instruct	5.25	23.76	11.77	56.20	64.53	60.30
RSUniVLM	10.85	28.95	18.96	53.27	61.42	54.65
TeoChat	8.72	20.14	13.27	32.11	45.39	36.40
EarthDial	7.13	24.43	17.84	51.10	59.83	51.80
QUAKE-CD	49.30	52.21	40.24	71.26	78.61	73.50

Table 2: Semantic evaluation on the reasoning trace of model outputs. *Human* annotators are given only the bi-temporal image pair and the model output.

by the hierarchical *QUAKE* syntax. In return, QUAKE-CD improves accumulated F1 over Text4Seg and RSUniVLM by 53.55% and 15.97%, respectively, establishing a stronger trade-off between throughput and spatial fidelity. The detailed throughput setup is provided in Appendix D.

Reasoning Quality Evaluation

Since mask metrics cannot verify whether textual claims stem from bi-temporal evidence, we evaluate the generated `<think>` and `<answer>` trajectories. Although we report BLEU-1, ROUGE-1, and METEOR for lexical overlap, Table 2 shows that generic VLMs score abnormally low under n -gram metrics, which penalize paraphrasing and verbose reasoning rather than semantic correctness. We therefore complement them with two semantics-aware automatic evaluators and a human-rating protocol. **Reranker Score** (Qwen3-Reranker-4B) assesses semantic fidelity, completeness, and structural coherence. **GPT-Judge Accuracy** (GPT-5.5-2026-04-23) quantifies semantic consistency against human-verified references (prompt templates in Figure 15 and 16). **Human Score** averages acceptance rates from two independent annotators over 1,000 randomly sampled outputs assessed for factual consistency and localization correctness, reaching 87.5% inter-annotator agreement.

Table 2 disentangles scaling from grounded bi-temporal supervision. Under traditional metrics, QUAKE-CD outperforms all baselines, achieving 49.30 BLEU-1 and 40.24 METEOR. On semantics-aware metrics, scaling Qwen3-VL from 8B to 235B raises the reranker score from 27.76 to 56.20 and GPT-judge accuracy from 47.78% to 64.53%, as greater capacity improves fluency and coarse semantic plausibility. Yet scale alone does not close the grounding gap: QUAKE-CD exceeds the same-8B baseline by 31.25% in human score and still leads the 235B model by 13.20%. RSUniVLM also trails QUAKE-CD by 18.85% in human score, indicating that its change-captioning supervision is insufficient for capturing the key bi-temporal evidence underlying each change. The agreement across lexical, semantic, and human evaluators indicates that the gain is robust rather than an artifact of any single judge.

Ablation Studies

Minimum Leaf Size. Figure 3 varies the minimum leaf size ℓ , which controls the trade-off between spatial resolution and sequence complexity, under fixed SFT settings. Reducing ℓ

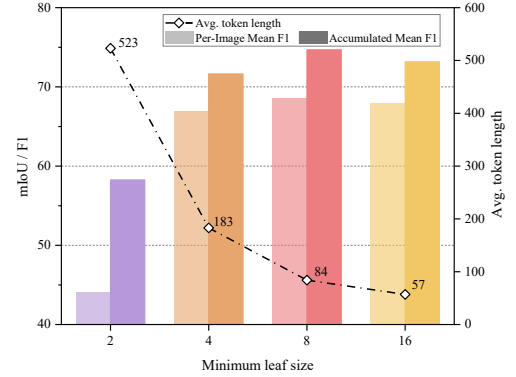


Figure 3: Ablation studies on the influence of the minimum leaf size ℓ .

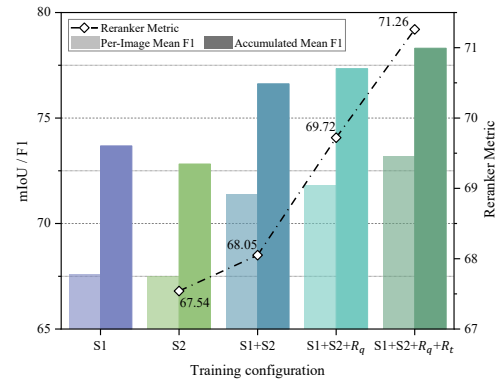


Figure 4: Ablation studies on the influence of curriculum stages and reward components.

from 16 to 2 increases the average sequence length from 57 to 523 tokens, yet mask quality does not improve monotonically. F1 reaches its maximum at $\ell=8$ and drops sharply at $\ell=2$. This degradation reflects a learnability bottleneck rather than a representational limit: although finer leaves can encode sharper boundaries, long recursive sequences increase grammar violations and dimension mismatches, weakening both SFT supervision and RL rollouts. At the opposite extreme, $\ell=16$ reduces the token budget but quantizes each 16×16 patch into a single bit, which erodes boundary precision. Thus, $\ell=8$ provides the strongest operating point, pre-

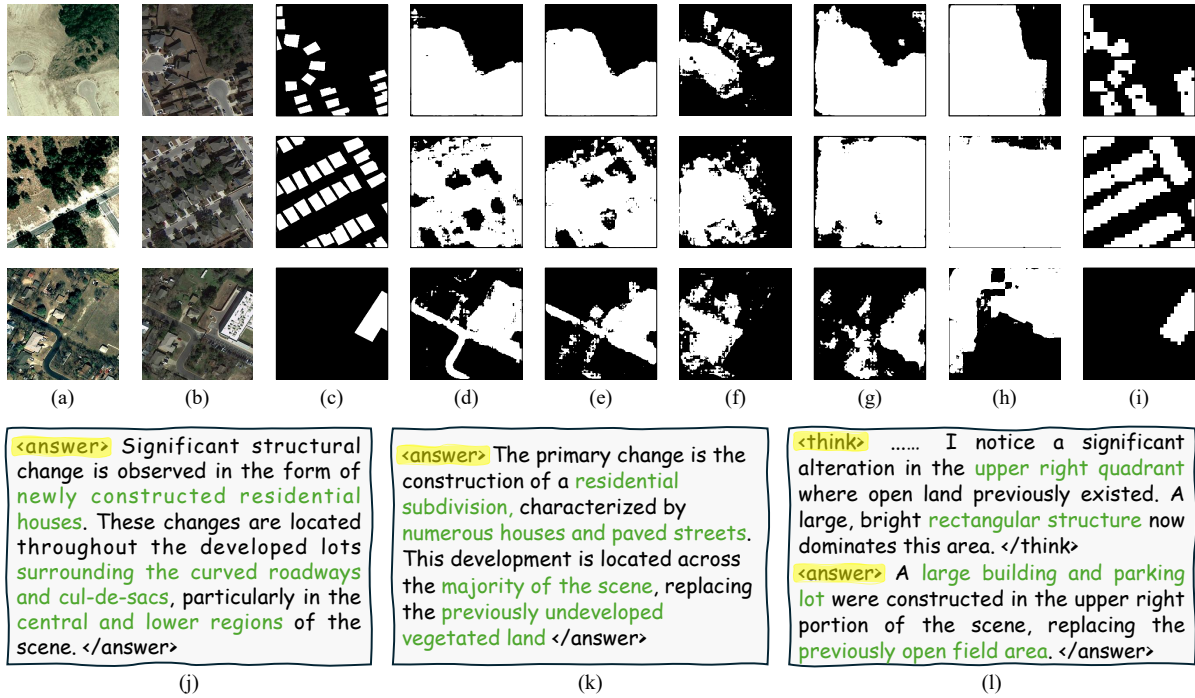


Figure 5: Qualitative results of representative decoder-based pixel-level VLMs and QUAKE-CD. (a): pre-change image. (b): post-change image. (c): ground truth. (d)-(i): results of LISA-7B, LISA-llama2-13B, GLaMM, PixelLM, Text4Seg, and QUAKE-CD. (j)-(l): textual change descriptions generated by QUAKE-CD.

serving fine-grained structure while keeping autoregressive decoding and policy optimization tractable.

Curriculum and Reward Components. Figure 4 evaluates five training regimes: S1, S2, S1+S2, S1+S2+ R_q , and S1+S2+ R_q + R_t . The S1 regime—supervising only the **<seg>** block—serves as the *QUAKE-only* variant that isolates the hierarchical representation from CoT supervision, while S1+S2 layers CoT atop the same syntax. First, the two-stage curriculum separates syntax acquisition from temporal reasoning: applying Stage 2 directly to the base VLM yields weaker per-image F1 due to interference between grammar learning and rationale generation, while initializing Stage-2 from the Stage-1 checkpoint improves both accumulated and per-image F1 by roughly 4% over either single-stage variant. Notably, the S1-only mask quality already exceeds every flat-serialization baseline in Table 1, attributing a substantial share of the dense-prediction gain to the QUAKE representation itself. Second, adding *Quadtree Reward* raises accumulated F1 by about 1% and the reranker score from 68.05 to 69.72, showing that grammar-gated spatial RL supplies a useful optimization signal beyond imitation learning. Third, the full dual-reward model reaches 71.26 reranker score with further gains in accumulated F1, confirming that *ThinkAnswer Reward* strengthens textual faithfulness without compromising the spatial accuracy preserved by *Quadtree Reward*, and, in turn, provides a stronger semantic guidance for QUAKE generation. Overall, the monotonic gains confirm that hierarchical encoding, staged supervision, and grammar-gated dual RL act synergistically across distinct failure modes.

Qualitative Analysis

Figure 5 presents representative cases where QUAKE-CD accurately resolves dense small-building construction requiring fine-grained bi-temporal comparison. External-decoder baselines and Text4Seg instead follow the high-recall pattern of Section 4, over-predicting broad regions in dense scenes and hallucinating large changed areas in single-building cases where they miss the pre- and post-change structural correspondence. In contrast, QUAKE-CD’s textual descriptions remain grounded, correctly specifying change type, location, and extent.

5 Conclusion

We introduced QUAKE-CD, a framework that formulates dense remote-sensing change detection as structured autoregressive generation under a syntax-verifiable mask grammar. Its quadtree mask language converts binary change maps into compact, grammar-constrained token sequences that can be decoded, checked, and optimized within autoregressive VLMs. With paired spatial supervision, grounded reasoning traces, staged curriculum learning, and grammar-gated dual-reward RL, QUAKE-CD aligns dense mask generation with faithful bi-temporal reasoning. Experiments show clear gains over decoder-based and flat text-as-mask VLM baselines, while demonstrating that grounded spatial supervision improves reasoning more effectively than generic model scaling. These results position structured spatial languages as a practical foundation for accurate, interpretable, and structurally checkable dense prediction in VLMs.

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A Quadtree Mask Encoding Details

This appendix extends the informal description in Section 3 with a complete formal specification of $\mathcal{L}_Q$, its deterministic encoder E , and decoder $\mathcal{D}$. The specification renders $\mathcal{L}_Q$ a syntactically and structurally checkable mask language whose well-formedness is decidable in linear time, independent of the model’s semantic predictions.

Formal Grammar of $\mathcal{L}_Q$

The alphabet of $\mathcal{L}_Q$ extends the base VLM vocabulary with 19 special tokens, supplemented by standard separators and numerals:

- **Structural delimiters:** $\langle \text{seg} \rangle$, $\langle / \text{seg} \rangle$
- **Uniform tile tokens:** $\langle \text{BG} \rangle$, $\langle \text{FG} \rangle$
- **Quadtree node tokens:** 15 nodes $\{\langle \text{Q0001} \rangle, \dots, \langle \text{Q1111} \rangle\}$; the all-zero pattern $\langle \text{Q0000} \rangle$ is strictly excluded, since an inactive quadrant is already absorbed by its parent node’s occupancy bit.
- **Separators and Modifiers:** $|$ (column delimiter), $\backslash n$ (row delimiter), and $*$ (run-length encoding prefix).

The syntax of $\mathcal{L}_Q$ is governed by a context-free grammar defined by the following production rules:

$$\text{Sequence} \rightarrow \langle \text{seg} \rangle \text{Grid} \langle / \text{seg} \rangle \quad (4)$$

$$\text{Grid} \rightarrow \text{Row} (\backslash n \text{Row})^* \quad (5)$$

$$\text{Row} \rightarrow \text{Tile} (| \text{Tile})^*$$

$$\text{Tile} \rightarrow \langle \text{BG} \rangle \mid \langle \text{FG} \rangle \mid \langle \text{BG} \rangle * n \mid \langle \text{FG} \rangle * n \mid \langle \text{Q}b_1b_2b_3b_4 \rangle \text{Children}(b_1b_2b_3b_4, d) \quad (6)$$

Here, $\text{Children}(b_1b_2b_3b_4, d)$ recursively expands every active quadrant with $b_i = 1$ in Z-order (preorder) traversal: top-left, top-right, bottom-left, bottom-right. Let d denote the current recursion depth, bounded by $d_{\max} = \lceil \log_2(32/\ell) \rceil$. Upon reaching $d_{\max}$, the quadrant collapses to a base Tile of side length ℓ .

Grammar validity ensures format compliance before decoding. A sequence is valid if and only if: (1) exactly one $\langle \text{seg} \rangle$ wrapper encloses the grid; (2) the expanded row and column counts match the expected grid dimensions $H/32 \times W/32$; (3) every run-length count satisfies $n \geq 2$ and does not exceed the remaining tiles in its row; (4) quadtree recursion depth never exceeds $d_{\max}$; and (5) the number of recursively expanded children matches the sum of active bits $\sum b_i$ in the corresponding $\langle \text{Q}abcd \rangle$ node.

Encoding Algorithm E

Algorithm 1 specifies the deterministic encoding procedure. Given a binary mask $M \in \{0, 1\}^{H \times W}$, a fixed tile size $T = 32$, and a minimum leaf size ℓ , the encoder partitions M into an $H/T \times W/T$ grid and recursively subdivides each heterogeneous tile until the minimum leaf size ℓ is reached.

The encoding is *recall-preserving* by construction: if $M[i, j] = 1$, then $\mathcal{D}(E(M))[i, j] = 1$. A single foreground pixel within any leaf forces that leaf’s occupancy bit to 1, and the decoder renders every active leaf as fully foreground. Precision loss therefore originates exclusively from boundary leaves of size ℓ , where mixed background pixels are

Algorithm 1: Quadtree Mask Encoding

Require: Binary mask $M \in \{0, 1\}^{H \times W}$, tile size $T = 32$, min leaf size $\ell = 8$

Ensure: Quadtree sequence $\mathbf{y}_{\text{seg}} \in \mathcal{L}_Q$

- 1: Partition M into grid $\mathcal{G}$ of $H/T \times W/T$ tiles of size $T \times T$
- 2: Initialize sequence buffer $S \leftarrow \langle \text{seg} \rangle$
- 3: **for** each row $r \in \mathcal{G}$ **do**
- 4: **for** each tile $t \in r$ **do**
- 5: **if** t is uniform background ($\forall_{i,j} t[i, j] = 0$) **then**
- 6: Emit $\langle \text{BG} \rangle$
- 7: **else if** t is uniform foreground ($\forall_{i,j} t[i, j] = 1$) **then**
- 8: Emit $\langle \text{FG} \rangle$
- 9: **else**
- 10: Emit $\text{ENCODENODE}(t, T, \ell, \text{depth} = 0)$
- 11: **end if**
- 12: Emit column delimiter $|$ if t is not the last tile in r
- 13: **end for**
- 14: Apply RLE: collapse consecutive identical uniform tokens to $\langle \text{BG} \rangle * n$ or $\langle \text{FG} \rangle * n$
- 15: Append row separator $\backslash n$
- 16: **end for**
- 17: Append $\langle / \text{seg} \rangle$
- 18: **return** S
- 19: **function** $\text{ENCODENODE}(B, \text{size}, \ell, \text{depth})$
- 20: $d_{\max} \leftarrow \lceil \log_2(\text{size}/\ell) \rceil$
- 21: **if** $\text{depth} = d_{\max}$ **then**
- 22: **return** $\langle \text{FG} \rangle$ if quadrant contains any foreground pixel
 else $\langle \text{BG} \rangle$ $\triangleright$ recall-preserving leaf rule
- 23: **end if**
- 24: $h \leftarrow \text{size}/2$
- 25: $b_1b_2b_3b_4 \leftarrow$ occupancy bits of quadrants TL, TR, BL, BR
- 26: **if** $b_1 = b_2 = b_3 = b_4 = 0$ **then** $\triangleright$ unreachable; defensive guard
- 27: **return** empty string
- 28: **end if**
- 29: Emit $\langle \text{Q}b_1b_2b_3b_4 \rangle$
- 30: **for** each quadrant q_i with $b_i = 1$ in TL, TR, BL, BR order **do**
- 31: $\text{ENCODENODE}(B[q_i], h, \ell, \text{depth} + 1)$
- 32: **end for**
- 33: **end function**

unavoidably promoted to foreground. For $\ell = 8$, the maximum false-positive expansion per boundary leaf is 64 pixels. This controlled precision–compression trade-off strictly eliminates false-negative boundary omissions, accounting for the bounded precision drop.

Total complexity is $O(HW)$: each pixel is visited once during partitioning, and recursive subdivisions operate on disjoint subregions.

Decoding Algorithm $\mathcal{D}$

Decoding reconstructs a binary mask $\hat{M}$ from a quadtree sequence $\mathbf{y}_{\text{seg}} \in \mathcal{L}_Q$ as a purely symbolic, non-parametric mapping, eliminating neural overhead during both inference and RL reward computation.

Upon extracting the $\langle \text{seg} \rangle \dots \langle / \text{seg} \rangle$ body, the decoder splits the grid by row separators, expands RLE prefixes, and validates the reconstructed sequence dimensions. Invalid sequences trigger categorical safety fallbacks:

Metric	Accumulated (%)	Per-Image Mean (%)
IoU	83.9	80.2
F1	91.2	88.0
Precision	83.9	80.2
Recall	100.0	100.0

Table 3: Round-trip encoding fidelity of E and D on the QUAKE-CoT training set.

ℓ	IoU	Accum. F1	Per-Image F1	Avg. Tokens
2	97.2	98.6	97.9	520.7
4	91.2	95.3	93.8	124.7
8	83.9	91.2	88.0	78.3
16	71.4	83.2	79.7	52.1

Table 4: Encoding fidelity by minimum leaf size ℓ on the QUAKE-CoT training set. Smaller leaves improve precision; larger leaves reduce sequence length.

- **Grammar Failure:** Missing wrappers, unrecognized node tokens, or malformed RLE prefixes yield an all-zero fallback mask.
- **Dimension Mismatch:** If expanded dimensions diverge from $H/32 \times W/32$, the decoder applies pad/crop alignment to recover a usable mask; the corresponding reward-side discount is specified in Appendix C.
- **Unexpected EOF:** Severely truncated sequences that break during token parsing default to the all-zero mask.
- **Depth Overflow:** Recursion exceeding $d_{\max}$ triggers an immediate fallback to a uniform $\ell \times \ell$ leaf-level approximation.

For valid sequences, tile tokens are rendered deterministically. $\langle BG \rangle$ and $\langle FG \rangle$ generate 32×32 zero or one blocks, respectively. A $\langle Qabcd \rangle$ node recursively renders only the quadrants whose occupancy bits $b_i = 1$, while inactive quadrants default to background. The rendered spatial blocks are concatenated in raster-scan order into the final dense mask $\hat{M} \in \{0, 1\}^{H \times W}$.

Encoding Fidelity and Compression Statistics

Table 3 quantifies the round-trip algorithmic fidelity of encoding followed by decoding across the QUAKE-CoT dataset. Foreground recall strictly achieves 100.0% at both accumulated and per-image levels, directly validating the claim of preserving all target foreground regions. Accumulated F1 of 91.2% and per-image F1 of 88.0% confirm that boundary-quantization precision loss remains strictly bounded under $\ell=8$. The decoded masks therefore serve as faithful supervision targets for SFT and as reliable reward signals for RL.

Table 4 ablates the minimum leaf size ℓ . Setting $\ell=2$ inflates sequence length, while $\ell=16$ shortens sequences at the cost of spatial resolution and a higher per-leaf false-positive rate. We adopt $\ell=8$ as the operating point that balances compression against precision.

B QUAKE-CoT Dataset Details

QUAKE-CoT pairs pixel-level change masks with chain-of-thought reasoning traces, supplying the linguistic supervision absent from purely discriminative change-detection benchmarks. We build upon established bi-temporal remote sensing benchmarks, augmenting them with hierarchical mask sequences and teacher-distilled reasoning traces.

Data Sources and Curation

We source raw image pairs and binary change masks from three large-scale change detection benchmarks: SYSU-CD (Shi et al. 2021), LEVIR-CD and LEVIR-CD+ (Chen and Shi 2020). Table 5 summarizes the characteristics of these source datasets. SYSU-CD covers diverse urban change types including buildings, roads, and vegetation, whereas the LEVIR series targets building additions and removals.

The curation process involves three stages:

Spatial Quantization. SYSU-CD natively provides 256×256 pairs and is consumed as-is. LEVIR-CD and LEVIR-CD+ (1024×1024) are partitioned into 16 non-overlapping 256×256 tiles per pair via raster scan. Each mask M is then encoded into y_{seg} via Algorithm 1 with $T=32$ and $\ell=8$. No positive-negative balancing is applied at this stage; sample-level filtering is deferred to the validation step described below.

Reasoning Distillation. We employ *Gemini-3.1-Pro-Thinking-Preview* (Team et al. 2023) as the teacher, conditioning generation on the bi-temporal pair (I_1, I_2) and the ground-truth mask M . The prompt in Figure 13 enforces two core constraints. First, the trace must reference only the two temporal images and pursue reasoning-oriented comparison—articulating how differences should be analyzed and which categories of change are relevant—rather than producing yes/no judgements. Second, the teacher must explicitly discount appearance shifts caused by season, phenology, sun angle, shadows, haze, or global brightness/contrast, focusing instead on structural change such as new or removed objects, altered built footprints, land-cover conversion, and layout reconfiguration. We issue 55,740 teacher calls in total; 9,788 (17.6%) require re-invocation due to API timeouts or format violations, and the affected samples are retained only after a successful re-call. Reasoning traces tokenized with the Qwen3-VL tokenizer average 84 tokens, with median 70 and range 31–303.

Validation. Four graduate students with backgrounds in remote sensing, computer vision, or computer science cross-verify all N_{trace} distilled traces against the corresponding bi-temporal pair and ground-truth mask, revising obvious factual inconsistencies. Traces judged factually consistent account for 98.7% of the corpus, while the remaining 1.3% are revised by the annotators. We therefore define the trace-validation rate as

$$A_{\text{trace}} = \frac{N_{\text{consistent}}}{N_{\text{trace}}} = 98.7\%. \quad (7)$$

We then apply three sample-level filters: (i) traces that remain factually inconsistent with the mask after revision are

Source	Resolution	Main change type	# Samples
SYSU-CD	256 × 256	Urban changes	20,000
LEVIR-CD	1,024 × 1,024	Building changes	637
LEVIR-CD+	1,024 × 1,024	Building and subtle changes	985

Table 5: Source datasets used to construct QUAKE-CoT. All original images were cropped or resized to 256 × 256 spatial resolution.

discarded; (ii) tiles whose change consists solely of sub-tile-scale boundary noise—small fragmented edge variations deemed too ambiguous for chain-of-thought training—are removed; (iii) to mitigate the heavy class imbalance toward unchanged tiles, we retain a uniformly random 20% of fully-empty-mask tiles. Filtering yields the final samples documented in Table 6.

Human Annotation and Agreement. All human assessments were conducted by the same four graduate annotators using a common rubric for factual consistency and spatial grounding. For the reported Human Score, we evaluated $N = 1,000$ randomly sampled outputs for each of the $M = 7$ evaluated models. Let $h_{m,i}^{(a)} \in \{0, 1\}$ denote the acceptance judgment assigned by annotator $a \in \{1, 2\}$ to the i -th output of model m . The acceptance rate of annotator a for model m is

$$A_m^{(a)} = \frac{1}{N} \sum_{i=1}^N h_{m,i}^{(a)}, \quad (8)$$

and the reported Human Score for model m averages the two annotators’ acceptance rates:

$$H_m = \frac{1}{2} \left(A_m^{(1)} + A_m^{(2)} \right) = \frac{1}{2N} \sum_{i=1}^N \sum_{a=1}^2 h_{m,i}^{(a)}. \quad (9)$$

Inter-annotator agreement is computed as exact label agreement over all MN evaluated outputs:

$$A_{\text{inter}} = \frac{1}{MN} \sum_{m=1}^M \sum_{i=1}^N \mathbb{1} \left[h_{m,i}^{(1)} = h_{m,i}^{(2)} \right] = 87.5\%. \quad (10)$$

To rule out the concern that the trained model merely re-plays teacher templates rather than grounding its reasoning in visual evidence, two of the same annotators additionally audit a stratified 500-sample subset of QUAKE-CD inference outputs, manually verifying that each rationale’s spatial claims—location, change type, and approximate extent—agree with the model’s own predicted mask. The rationale-mask agreement rate is defined as

$$A_{\text{rm}} = \frac{N_{\text{mask-consistent}}}{500} = 97.2\%, \quad (11)$$

indicating that the textual reasoning is internally consistent with the spatial prediction rather than decoupled boilerplate.

Data Partitioning

The dataset is partitioned into training and testing sets following the splits defined in the source benchmarks, enabling direct comparison with prior change-detection methods. Table 6 details the final data splits. The test split is held out for the final evaluation of both the SFT baselines and the GRPO-trained QUAKE-CD model.

Split	Size	Purpose
Train	22,007	Stage 1 & Stage 2
Train	5,502	RM training & GRPO
Test	9,301	Final evaluation

Table 6: QUAKE-CoT data partition. The dataset is divided into a training split and a held-out test split, maintaining the original dataset distributions.

Dataset Statistics

The final QUAKE-CoT corpus comprises 36,810 aligned (image pair, mask, reasoning trace) triples drawn from the three source benchmarks. Reasoning traces total approximately 3.09M Qwen3-VL tokens, averaging 60.4 words (84 tokens) per trace—roughly an order of magnitude longer than the single-sentence captions in LEVIR-CC (Liu et al. 2022) (7.99 words) and DUBAI-CC (Hoxha et al. 2022) (7.35 words). QUAKE-CoT is also the only corpus among the compared datasets to pair these reasoning traces with deterministically decodable dense mask sequences.

Comparison with Existing Benchmarks

Table 7 compares QUAKE-CoT with representative datasets across change detection and change captioning paradigms. Standard discriminative benchmarks such as LEVIR-CD and SYSU-CD provide only binary masks, leaving language supervision to downstream users. Existing change-captioning datasets like LEVIR-CC (Liu et al. 2022) and DUBAI-CC (Hoxha et al. 2022) attach a single short caption per pair, which seldom encodes the spatial localization or step-wise comparison needed for chain-of-thought training. QUAKE-CoT couples each pair with a multi-step reasoning trace and a deterministically decodable dense mask, supporting joint optimization of language and spatial grounding within a single autoregressive objective.

Figure 6 visualizes the lexical distribution of QUAKE-CoT’s reasoning traces. The vocabulary is dominated by structural objects (*building*, *landcover*) and change descriptors (*new*, *modification*, *conversion*), reflecting the spatial-temporal vocabulary required for bi-temporal reasoning rather than the surface-level descriptors typical of single-sentence captioning.

Licenses and Limitations

QUAKE-CoT is released under the licenses of its underlying source datasets (SYSU-CD, LEVIR-CD, and LEVIR-CD+); we redistribute the bi-temporal imagery together with the

structure is preserved verbatim; (iii) giveaway tokens such as “mask” or “ground truth” are forbidden.

y_{neg} is therefore placed at the bottom of the preference hierarchy defined below.

Preference Pair Construction

From the $K = 10$ rollouts we select three samples spanning the IoU spectrum:

$$y_a = \arg \max_{y_i} \text{IoU}(y_i), \quad (13)$$

$$y_b = \arg \min_{y_i} \text{IoU}(y_i), \quad (14)$$

$$y_c = \arg \max_{y_i \notin \{y_a, y_b\}} |\text{IoU}(y_a) - \text{IoU}(y_i)| \cdot |\text{IoU}(y_i) - \text{IoU}(y_b)|. \quad (15)$$

Here y_a and y_b are the highest- and lowest-IoU rollouts, while y_c is the intermediate rollout whose IoU sits farthest from both endpoints, measured by the product of absolute IoU gaps.

A preference comparison between any two candidates (y_i, y_j) is considered valid only if their performance discrepancy is significant enough to warrant a discernible semantic difference. We enforce this through a margin check threshold $\tau = 0.1$:

$$y_i \succ y_j \iff \text{IoU}(y_i) - \text{IoU}(y_j) \geq \tau \quad (16)$$

By incorporating the ground-truth sequence y_{gt} (considered the absolute upper bound) and the fabricated hard negative y_{neg} (the absolute lower bound), we establish a stringent, multi-tiered preference hierarchy:

$$y_{\text{gt}} \succ y_a \succ y_c \succ y_b \succ y_{\text{neg}} \quad (17)$$

From this hierarchy, we extract the preference set

$$\mathcal{D}_{\text{pref}} = \{(y_c, y_r) \mid y_c \succ y_r\}, \quad (18)$$

instantiated by pairs such as $(y_{\text{gt}}, y_{\text{neg}})$, (y_b, y_{neg}) , (y_a, y_c) , and (y_c, y_b) , provided the τ gap condition is satisfied for rollout comparisons. These pairs are aggregated to train the Qwen3-VL-8B-Instruct reward model. The neural reward model is optimized using the following objective:

$$\mathcal{L}_{\text{RM}} = -\mathbb{E}_{\mathcal{D}_{\text{pref}}} [\log \sigma(R_\phi(y_c) - R_\phi(y_r)) - \lambda(R_\phi(y_c) + R_\phi(y_r))^2], \quad (19)$$

The resulting preference set anchors R_ϕ to spatial fidelity rather than surface phrasing.

Dual-Reward Specification

During the GRPO RL stage, the total reward combines a rule-based R_q with the learned R_t :

$$R(\mathbf{x}, y) = (1 - \mu) R_q(y; M) + \mu R_t(y; \mathbf{x}), \quad (20)$$

with $\mu = 0.1$. The rule-based component validates grammar and spatial accuracy; the learned component scores the bi-temporal reasoning trace.

Quadtree Reward Formulation (R_q). The Quadtree Reward deterministically validates format compliance and spatial accuracy:

$$R_q = \text{clip}(R_{\text{struct}} + R_{\text{content}}, 0.0, 1.0) \quad (21)$$

If the generation fails $\mathcal{L}_Q$ parsing—missing `<seg>` wrapper, unrecognized tokens, a forbidden `<Q0000>` node, an RLE count below 2, or a child count not matching $\sum_i b_i$ for any `<Qabcd>` node—grammar gating forces $R_q = 0.0$. Otherwise:

- **Structure Score (R_{struct}):** The decoded grid is compared against the expected dimensions (H_e, W_e) via row-wise error rates and an exact-alignment bonus:

$$\epsilon_h = \min(1, |H_p - H_e|/H_e), \quad (22)$$

$$\epsilon_w = \frac{1}{H_e} \sum_{i=1}^{H_e} \min(1, |W_i - W_e|/W_e), \quad (23)$$

$$R_{\text{struct}} = 0.25 \max(0, 1 - \frac{1}{2}(\epsilon_h + \epsilon_w)) + 0.05 \cdot \mathbb{I}[H_p = H_e \wedge \forall i W_i = W_e], \quad (24)$$

where H_p is the decoded row count and W_i the logical width of row i after RLE expansion.

- **Content Score (R_{content}):** The decoded binary mask $\hat{M}$ is scored against the ground truth M via the Tversky index $\mathcal{T}_{\alpha, \beta}$ with $(\alpha, \beta) = (0.3, 0.7)$, which weights false negatives more strongly than false positives. Writing $\text{TP} = |M \cap \hat{M}|$, $\text{FP} = |\hat{M} \setminus M|$, and $\text{FN} = |M \setminus \hat{M}|$,

$$\mathcal{T}_{\alpha, \beta}(M, \hat{M}) = \frac{\text{TP}}{\text{TP} + \alpha \text{FP} + \beta \text{FN}}. \quad (25)$$

The content score is non-trivial only when the ground-truth mask contains foreground: when $|M| = 0$, an exact-empty prediction earns a residual reward of 0.2 to prevent collapse to all-background outputs, while any false-positive prediction yields 0. For active masks ($|M| > 0$):

$$R_{\text{content}} = 0.70 \mathcal{T}_{\alpha, \beta}(M, \hat{M}) s_{\text{align}} + 0.03 \mathbb{I}[\mathcal{T} \geq 0.45 \wedge \text{strict}], \quad (26)$$

where $s_{\text{align}} = 1.0$ under strict decode and $s_{\text{align}} = 0.80$ when the pad/crop fallback was triggered. The additive +0.03 rewards strict-decoded predictions whose Tversky score clears the boundary-preservation threshold.

ThinkAnswer Reward Formulation (R_t). The reward model R_ϕ trained in the preceding RM stage scores the extracted `<think>...</think><answer>...</answer>` segment of each rollout. To stabilise the reward scale across queries, we apply query-wise centred sigmoid normalisation over the group $\mathcal{S}_{\mathbf{x}}$ of $K = 8$ rollouts sharing prompt $\mathbf{x}$:

$$R_t(y_i) = \sigma(R_\phi(y_i | \mathbf{x}) - \bar{R}_\phi(\mathbf{x})), \quad (27)$$

$$\bar{R}_\phi(\mathbf{x}) = \frac{1}{|\mathcal{S}_{\mathbf{x}}|} \sum_{y_j \in \mathcal{S}_{\mathbf{x}}} R_\phi(y_j | \mathbf{x}). \quad (28)$$

The mean is computed across all K rollouts of $\mathbf{x}$ aggregated over all data-parallel ranks, so every rank applies identical centring; this bounds $R_t \in (0, 1)$ and yields zero-mean advantages within each query group.

Hyperparameter	Stage 1 (SFT)	Stage 2 (SFT)	RM Training	RL (GRPO)
Foundation Model	Qwen3-VL-8B	Stage 1 Checkpoint	Qwen3-VL-8B	Stage 2 Checkpoint
Training Epochs	10	5	3	3
Optimizer	AdamW	AdamW	AdamW	AdamW
Precision	BF16	BF16	BF16	BF16
Learning Rate	2×10^{-4}	2×10^{-4}	1×10^{-4}	5×10^{-6}
LR Scheduler	Cosine	Cosine	Cosine	Cosine
Warmup Ratio	0.03	0.03	0.05	0.03
Weight Decay	0.01	0.01	0.01	0.01
Hardware Setup	8× H200	8× H200	4× H200	8× H200
Batch Size (BS) per GPU	8	8	4	2
Gradient Accumulation Steps	1	1	1	1
Max Sequence Length	4096	4096	2048	1024 (Completion)
<i>LoRA Configuration</i>				
LoRA Target Modules	Attention & MLP	Attention & MLP	Attention & MLP	All Linear
LoRA Rank (r)	64	64	64	32
LoRA Alpha (α)	128	128	128	64
LoRA Dropout	0.05	0.05	0.05	0.05
<i>RL-Specific Configuration</i>				
Rollouts per Prompt (K)	—	—	—	8
Temperature	—	—	—	0.7
KL Coefficient (β)	—	—	—	0.04
Reward Weights	—	—	—	1.0 (R_q), 0.1 (R_t)

Table 8: Comprehensive hyperparameters for all training stages of QUAKE-CD. BS = batch size per device.

D Comprehensive Hyperparameters

Table 8 lists the full hyperparameter configuration for the two SFT stages, the reward model, and the GRPO rollout.

Throughput Measurement Protocol. Because Table 1 mixes specialist change detectors and VLMs, we define throughput at the application level rather than the token level. All methods are measured on a single NVIDIA H200 GPU with batch size 64 under BF16 inference, and the reported numbers are wall-clock end-to-end throughput in pairs/s. For specialist detectors, this includes direct mask prediction and deterministic post-processing. For VLM baselines, it includes prompt encoding, autoregressive generation until the model’s stopping criterion, and final mask extraction when applicable. For QUAKE-CD, this additionally includes generation of the `<seg>` span, followed by deterministic decoding of `<seg>` into the binary mask. No test-time adaptation, model parallelism, or extra refinement is used.

E More Experimental Results

Oracle Performance Upper Bound

The mask representation imposes an upper bound independently of model generation. Table 9 reports the representation oracles of the hierarchical *QUAKE* encoding and the flat text-as-mask encoding on the QUAKE-CoT test set. The flat oracle applies to Text4Seg and RSUniVLM, while the *QUAKE* oracle applies to QUAKE-CD.

Table 9 isolates the effect of mask representation. Both encodings preserve 100.00% foreground recall, confirming

that neither representation discards annotated change regions. Compared with flat text-as-mask encoding, *QUAKE* increases accumulated precision from 68.04% to 84.03% and per-image precision from 65.23% to 81.43%. These improvements raise accumulated and per-image F1 by 10.35% and 9.70%, respectively. The results show that hierarchical refinement suppresses boundary over-coverage induced by uniform spatial quantization while preserving complete foreground coverage.

For metric $m \in \{\text{Precision, Recall, F1}\}$ and evaluation regime $s \in \{\text{acc, img}\}$, the fraction of oracle performance reached is

$$\rho_m^{(s)} = \frac{m_{\text{model}}^{(s)}}{m_{\text{oracle}}^{(s)}} \times 100\%, \quad (29)$$

where $m_{\text{oracle}}^{(s)}$ is the upper bound obtained from representation round-trip fidelity and $m_{\text{model}}^{(s)}$ is the end-to-end result in Table 1. These ratios quantify the remaining model-generation gap after accounting for representation-specific quantization.

Table 10 shows that QUAKE-CD realizes the largest fraction of its representation oracle across all metrics. It reaches 85.75% accumulated F1 and 82.52% per-image F1, exceeding the strongest alternative by 8.76% and 5.35%, respectively. Its normalized precision also remains above 84% under both evaluation regimes, indicating substantially less degradation from autoregressive generation. Together, the two tables show that the gains of QUAKE-CD arise from both a more faithful spatial representation and a more effec-

Representation	Accumulated Mean			Per-Image Mean		
	Precision	Recall	F1	Precision	Recall	F1
Flat text-as-mask	68.04	100.00	80.97	65.23	100.00	78.98
QUAKE	84.03	100.00	91.32	81.43	100.00	88.68

Table 9: Representation-oracle mask quality on the QUAKE-CoT test set. Each ground-truth mask is encoded with the corresponding mask representation and deterministically decoded without model generation. All metrics are reported as percentages.

Method	Accumulated Mean			Per-Image Mean		
	Precision	Recall	F1	Precision	Recall	F1
Text4Seg	21.34	84.07	30.58	23.41	54.88	24.88
RSUniVLM	73.67	82.45	76.99	74.74	81.23	77.17
QUAKE-CD	84.48	87.31	85.75	84.45	84.09	82.52

Table 10: End-to-end mask quality normalized by the corresponding representation oracle on the QUAKE-CoT test set. Each entry reports model performance divided by oracle performance.

Dataset	Accumulated Mean			Per-Image Mean		
	Precision	Recall	F1	Precision	Recall	F1
LEVIR-CD	64.65	98.61	78.10	58.53	90.23	71.00
LEVIR-CD+	55.79	85.82	67.62	57.23	78.68	66.26

Table 11: Change-mask performance under tiled inference on the original 1024×1024 test images. All values are percentages.

tive realization of its attainable mask quality.

Scalability to High-Resolution Imagery

Although our model is trained with 256×256 image pairs, practical remote sensing imagery often covers substantially larger spatial extents. We therefore evaluate its scalability on the original 1024×1024 images from LEVIR-CD and LEVIR-CD+, without high-resolution fine-tuning or architectural modification. Each co-registered bi-temporal pair is partitioned into a 4×4 grid of non-overlapping and spatially aligned 256×256 tile pairs. The model processes each pair independently at its native training resolution. The resulting *QUAKE* predictions are decoded into binary tile masks, restored to their original coordinates, and stitched into a full-resolution 1024×1024 prediction for evaluation against the original ground truth. This protocol preserves local image detail and avoids the information loss caused by resizing the entire image to the training resolution. No overlapping inference, test-time adaptation, cross-tile refinement, or learned fusion is employed.

Table 11 reports the resulting full-resolution performance. On the test pairs of LEVIR-CD, the model achieves a dataset-level precision of 64.65% and an F1 score of 78.10%, with a recall of 98.61%. On the test pairs of LEVIR-CD+, it obtains a precision of 55.79% and an F1 score of 67.62%, with a recall of 85.82%. The consistently high recall indicates that the model retains strong sensitivity to changed structures when applied to images with $16\times$ the training image area. The lower precision on LEVIR-CD+ suggests that the remaining errors are primarily attributable to conservative over-detection rather than missed changes. Overall, these results demonstrate that the proposed *QUAKE* predictor can

be effectively extended to megapixel-scale imagery through a simple tiled inference strategy while preserving the original training resolution.

Figure 7 provides qualitative evidence consistent with these results. In the first example, the stitched prediction recovers the full extent of the newly developed curvilinear residential area, while merging nearby building footprints and expanding into some surrounding regions. In the second example, it captures the long rows of newly constructed buildings but represents several adjacent instances as contiguous bands. These errors explain the high recall and lower precision reported in Table 11. Despite independent tile processing, both predictions retain the global spatial arrangement of changes after stitching, confirming effective transfer from 256×256 training crops to 1024×1024 scenes.

Generalization on Out-of-Domain Dataset

All preceding evaluations use data drawn from QUAKE-CoT and therefore retain substantial overlap with its training distribution. We assess external validity on WHU Building CD (Ji, Wei, and Lu 2019), which is excluded from data construction, supervised fine-tuning, and reinforcement learning. WHU has a ground sampling distance of 0.2 m/pixel, compared with 0.3 m/pixel for LEVIR-CD and 0.5 m/pixel for SYSU-CD, and was acquired with a different sensor. This benchmark consequently introduces coupled shifts in spatial resolution, sensor characteristics, scene appearance, and object scale.

We evaluate the frozen QUAKE-CD model without specific fine-tuning or test-time adaptation. The native ground sampling distance is preserved, and each pair is processed using the same tiled inference and deterministic decoding

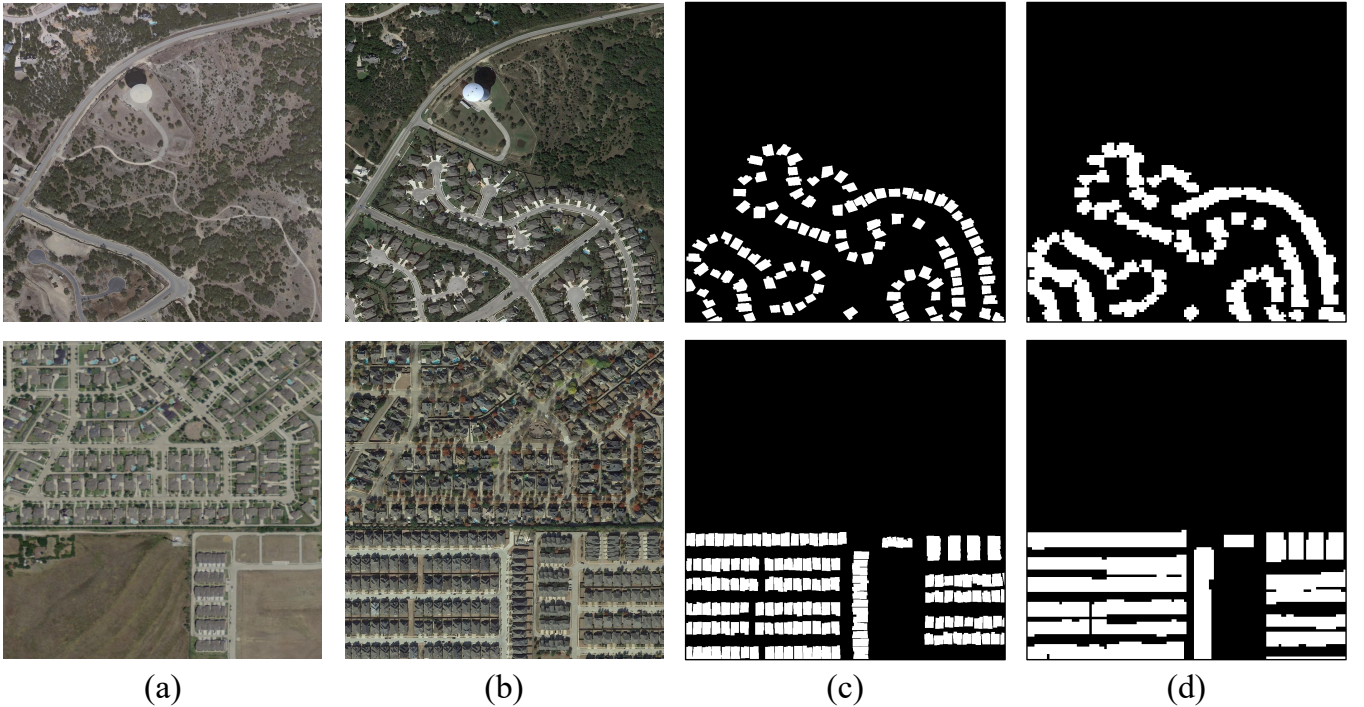


Figure 7: Qualitative results on original 1024×1024 image pairs under tiled inference. Each row presents one example: (a) pre-change image; (b) post-change image; (c) ground-truth change mask; (d) stitched prediction.

Method	Accumulated Mean			Per-Image Mean		
	Precision	Recall	F1	Precision	Recall	F1
BIT	72.93	68.25	68.54	62.24	60.44	60.21
ChangeFormer	75.83	71.04	75.36	71.30	60.63	65.29
ChangeCLIP	76.41	68.70	73.09	54.43	45.76	47.13
LISA-7B	15.73	64.03	25.96	16.46	42.15	19.97
LISA-llama2-13B	14.88	68.52	22.99	17.62	41.71	21.15
GLaMM	12.34	45.36	23.00	14.23	27.53	15.93
PixelLM	16.02	65.63	26.75	17.72	42.73	20.87
Text4Seg	14.65	74.41	22.26	12.94	47.80	19.18
RSUniVLM	45.23	74.57	58.25	46.80	72.62	56.65
QUAKE-CD	65.36	78.40	71.29	64.90	73.88	69.10
QUAKE-CD / Oracle	77.84	78.40	78.10	87.69	73.88	82.44

Table 12: Out-of-domain performance on WHU Building CD. QUAKE-CD is evaluated without specific fine-tuning. The second row reports end-to-end performance normalized by the corresponding representation oracle.

protocol described in Section 4.

Table 12 reports the end-to-end mask quality of QUAKE-CD and the fraction of the corresponding representation oracle attained by the model. The oracle is obtained by encoding and deterministically decoding the WHU ground-truth masks with the *QUAKE* representation. The normalized row follows Equation 29 and divides each end-to-end metric by its oracle counterpart. Under accumulated evaluation, QUAKE-CD remains competitive with the specialist change detectors: its accumulated F1 of 71.29% is within 4.07% of the strongest specialist, ChangeFormer, while its accumulated recall of

78.40% is the best in the table. More importantly, QUAKE-CD surpasses all specialist baselines in the per-image setting, reaching 69.10% F1 versus 65.29% for the strongest specialist, ChangeFormer. Among the VLM baselines, QUAKE-CD is the best by a clear margin; the strongest VLM baseline, RSUniVLM, reaches only 58.25% accumulated F1 and 56.65% per-image F1, leaving QUAKE-CD ahead by 13.04 and 12.45 points, respectively. Without adaptation, QUAKE-CD achieves accumulated and per-image F1 scores of 71.29% and 69.10%. These results correspond to 78.10% and 82.44% of the respective oracle F1 scores. Recall remains higher

(α, β)	Accumulated Mean			Per-Image Mean		
	Precision	Recall	F1	Precision	Recall	F1
(0.1, 0.9)	70.82	86.68	77.95	68.41	83.04	72.63
(0.2, 0.8)	70.24	86.89	77.68	68.19	83.24	72.51
(0.3, 0.7)	70.99	87.31	78.31	68.77	84.09	73.18
(0.4, 0.6)	73.13	81.19	76.95	69.63	79.44	71.61
(0.5, 0.5)	73.09	81.31	76.98	69.40	79.35	71.46

Table 13: Sensitivity of the Tversky parameters in the Quadtree Reward.

μ	Accumulated Mean			Per-Image Mean			Reranker
	Precision	Recall	F1	Precision	Recall	F1	(score)
0.05	68.91	76.38	72.45	58.70	74.50	67.50	69.72
0.10	70.99	87.31	78.31	68.77	84.09	73.18	71.26
0.15	70.55	86.75	77.82	68.31	83.17	72.60	70.30
0.20	73.09	81.31	76.98	69.40	79.35	71.46	68.05

Table 14: Sensitivity of the reward trade-off coefficient in Eq. (3).

KL Coefficient	Accumulated Mean			Per-Image Mean			Reranker
	Precision	Recall	F1	Precision	Recall	F1	(score)
0.00	72.78	81.53	76.90	69.44	79.67	71.58	64.85
0.02	71.60	82.91	76.84	68.77	80.66	71.64	70.82
0.04	70.99	87.31	78.31	68.77	84.09	73.18	71.26
0.06	68.91	76.38	72.45	66.03	74.50	67.50	70.79
0.08	69.25	76.31	72.61	66.12	74.46	67.48	64.09

Table 15: Sensitivity of the GRPO KL coefficient. The Reranker column reports the average reranker score.

than precision under both evaluation regimes, showing that the transferred model favors coverage of changed buildings over conservative boundary estimation. The results establish effective generalization across shifts in spatial resolution, sensor characteristics, and scene appearance.

Qualitative Attention Visualization

To assess the spatial grounding of the structured prediction process, we construct two complementary attention views. The **token-conditioned visualization** projects the attention associated with each generated *QUAKE* token onto its designated visual tile. The **aggregated overlay** pools the visual responses over the complete segmentation sequence, providing a holistic view of the evidence supporting the predicted change region.

We present three representative examples in Figure 8, Figure 9, and Figure 10. Across all cases, both attention views concentrate on regions undergoing meaningful temporal changes, while largely unchanged background areas receive weak responses. In the token-conditioned visualization, the red box marks the visual tile corresponding to the generated *QUAKE* token. The peak attention mostly falls within this tile, with substantially lower responses on neighboring and non-corresponding tiles. This self-tile concentration provides direct evidence of the spatial alignment between the *QUAKE* output and the visual representation. The aggregated attention overlay further recovers the overall extent of

the changed regions and follows their salient structures and boundaries. Its spatial distribution is consistent with both the predicted masks and the ground-truth annotations, indicating that the model focuses on relevant changes while suppressing irrelevant contextual content. This agreement is maintained across examples with different object scales, shapes, and background configurations, supporting the effectiveness of the *QUAKE*-based representation for grounding structured segmentation predictions in visual change evidence.

Parameter Sensitivity Analysis

We sweep three reward-related hyperparameters around the default configuration used in the main experiments: the Tversky asymmetry $(\alpha, \beta) = (0.3, 0.7)$, the reward trade-off $\mu = 0.1$ in Eq. (3), and the KL coefficient 0.04. As summarized in Tables 13–15, the default setting is consistently the most balanced choice: it yields the strongest recall/F1 trade-off for the Tversky reward, achieves the highest reranker score at $\mu = 0.1$, and provides the best overall balance under KL regularization. Although some nearby settings improve precision in isolation, they generally do so at the expense of recall, F1, or reranker performance.

F Failure Mode Analysis

We quantify three categories of generation failure that can compromise mask reconstruction. **Grammar invalidity** oc-

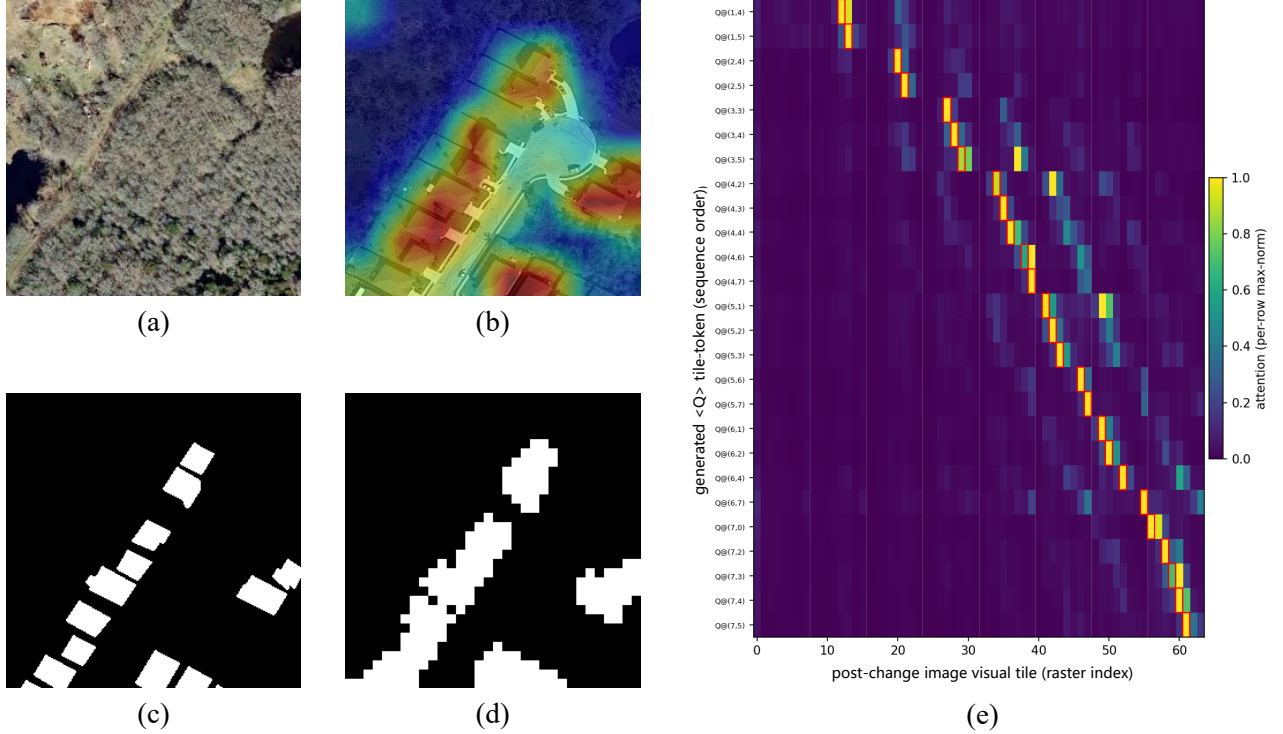


Figure 8: Qualitative attention visualization example. (a) Pre-change image; (b) post-change image with aggregated attention overlay for the complete segmentation sequence; (c) ground-truth change mask; (d) predicted change mask; (e) token-conditioned attention visualization, where the red box indicates the visual tile corresponding to the generated *QUAKE* token. The token-conditioned attention peaks within the corresponding red-boxed tile, while the aggregated attention concentrates on regions of temporal change and is consistent with both the predicted and ground-truth masks.

Stage	Grammar Invalid	Structure Mismatch / Fallback
After Curriculum SFT	1.26%	12.64%
After RL	0.13%	2.17%

Table 16: Failure-mode rates on the *QUAKE*-CoT test split. The pad/crop fallback is invoked exactly when a structure mismatch is detected, so the two columns share a trigger.

curs when the $\langle \text{seg} \rangle$ span cannot be parsed under $\mathcal{G}_Q$ (e.g., malformed quadtree node, run-length count below 2, or unmatched delimiters). **Structure mismatch** occurs when the parsed grid does not match the expected $H/32 \times W/32$ dimensions. Whenever a structure mismatch is detected, the decoder falls back to a **pad/crop alignment** that pads or truncates the recovered grid to the target shape; the trigger rate of this fallback therefore coincides with the structure-mismatch rate.

Table 16 reports these rates measured on the held-out test split after the two-stage curriculum ($S1+S2$) and after grammar-gated dual-reward RL ($S1+S2+R_q+R_t$). Curriculum supervision already keeps grammar invalidity at 1.26%, indicating that the syntax is largely internalized during SFT. Reinforcement learning further reduces grammar invalidity to 0.13%, an order-of-magnitude drop attributable to the grammar gate in R_q that assigns zero reward to ungrammatical rollouts. Structure mismatch decreases from 12.64%

to 2.17% over the same transition, showing that the structure score in R_q effectively penalizes grid-dimension drift even though the grammar gate alone does not enforce it.

G Prompt Templates

Figures 11–16 reproduce every prompt used in this work. Data construction prompts (Figs. 11, 12, 13, 14) cover the system instruction, QA distillation, and hard-negative synthesis described in §B and §C. Evaluation prompts (Figs. 15, 16) cover the reranker used during preference-pair selection and the GPT judge used in §4.

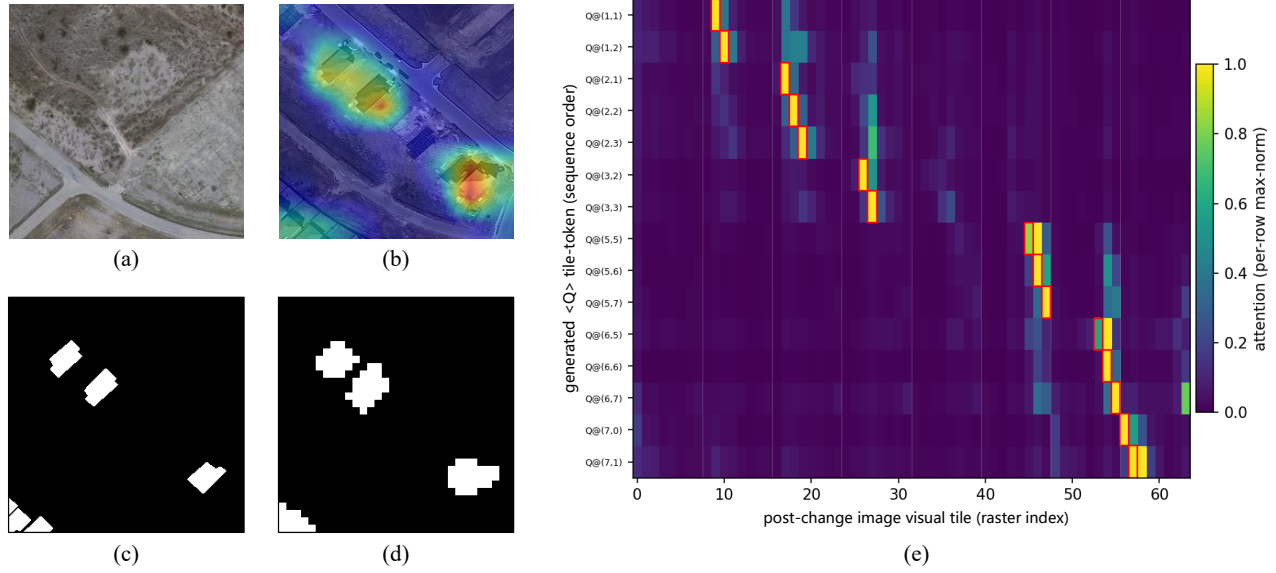


Figure 9: Qualitative attention visualization example. (a) Pre-change image; (b) post-change image with aggregated attention overlay for the complete segmentation sequence; (c) ground-truth change mask; (d) predicted change mask; (e) token-conditioned attention visualization, where the red box indicates the visual tile corresponding to the generated *QUAKE* token. The token-conditioned attention peaks within the corresponding red-boxed tile, while the aggregated attention concentrates on regions of temporal change and is consistent with both the predicted and ground-truth masks.

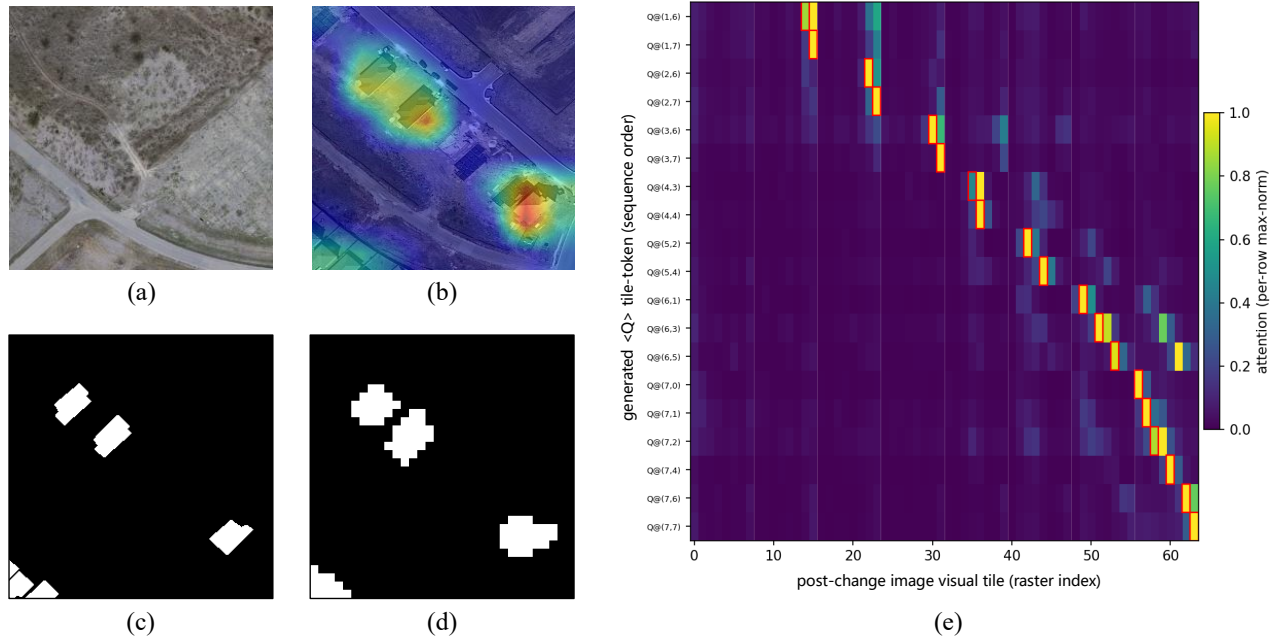


Figure 10: Qualitative attention visualization example. (a) Pre-change image; (b) post-change image with aggregated attention overlay for the complete segmentation sequence; (c) ground-truth change mask; (d) predicted change mask; (e) token-conditioned attention visualization, where the red box indicates the visual tile corresponding to the generated *QUAKE* token. The token-conditioned attention peaks within the corresponding red-boxed tile, while the aggregated attention concentrates on regions of temporal change and is consistent with both the predicted and ground-truth masks.

System Prompt

You are an expert in remote sensing change detection.

You are given:

- two co-registered temporal images
- a user question about changes between them

Your job is to analyze the changes and produce exactly three tagged blocks in this exact order:

<think>...</think>

<answer>...</answer>

<seg>...</seg>

Output contract:

- Output exactly these three tagged blocks, in the exact order above.
- Do not output any extra text before, after, or between the blocks.
- Do not use markdown fences.
- Do not add titles, explanations, or notes outside the three tags.

Tag requirements:

- <think>: brief visual reasoning about the changes between the two images.
- <answer>: direct and concise final answer to the user question.
- <seg>: only the encoded change mask, with no explanations and no extra characters.

Important:

- If there is no detected change, <seg> should still contain a valid encoding using the same format.
- If natural-language preferences conflict with the mask encoding rules, follow the mask encoding rules.

<QUAKE Rules>

Figure 11: System prompt prepended to all training and inference inputs of QUAKE-CD.

QUAKE Rules (Embedded in System Prompt)

Encoding specification for <seg>:

Overall layout:

- Partition the image into a regular grid of non-overlapping 32×32 pixel tiles.
- Tiles are ordered in raster order: top-to-bottom by 32-pixel-high bands, and within each band left-to-right.
- One text line corresponds to one 32-pixel-high band.
- Within a line, adjacent tile encodings are separated by a single "|" with no required spaces.
- The k-th text line (0-based) covers image rows $[k \times 32, (k+1) \times 32)$.
- Within that line, the t-th tile position after expanding any run-length tokens (0-based) covers image columns $[t \times 32, (t+1) \times 32)$.
- Therefore, the number of text lines is $\text{image_height} / 32$, and each line represents $\text{image_width} / 32$ tiles after run-length expansion.

Valid tile encodings:

Each tile must be encoded as exactly one of the following:

- "<BG>"
meaning the entire 32×32 tile is background.
- "<FG>"
meaning the entire 32×32 tile is foreground at the target output granularity.
- "<BG>*n" or "<FG>*n"
meaning a run-length encoding for consecutive identical "<BG>" or "<FG>" tiles on the same line only.
Here n is an integer ≥ 2 .
Run-length compression applies only within a single line, never across lines.
- A quadtree string
meaning a concatenation of one or more blocks of the form "<Qabcd>" with no spaces.

Quadtree bit semantics:

- In each "<Qabcd>" block, the bits a, b, c, d correspond to quadrants in this exact order:
top-left, top-right, bottom-left, bottom-right.
- Bit meaning:
 - 1 = this quadrant contains foreground / is active
 - 0 = this quadrant is empty / inactive

Quadtree serialization rules:

- A quadtree string is serialized in preorder.
- Write the current "<Qabcd>" first.
- Then, for each quadrant whose bit is 1, recursively append that child quadrant's own "<Q....>" block(s).
- The recursive child order must be exactly:
top-left, top-right, bottom-left, bottom-right.
- Do not append child blocks for quadrants whose bit is 0.
- Recursion continues until the minimum leaf size reaches 8×8 .
- At the final level, one "<Qabcd>" describes the four 8×8 sub-quadrants of its parent region.

Final rules for <seg>:

- <seg> must contain only the encoded mask.
- Do not include prose, comments, labels, explanations, markdown, or any characters outside this encoding.

Figure 12: QUAKE grammar rules embedded into the system prompt to constrain the model's output schema.

Prompt for QUAKE-CD-COT Generation

You are an expert in remote sensing change detection.

The user sends three co-registered remote sensing images, in order: (1) Time-1 (earlier), (2) Time-2 (later), (3) a binary change reference (white = changed, black = unchanged). Use (3) only to keep outputs **internally consistent** with real change locations; **never mention, cite, or allude to (3)** in the JSON text.

Return **one valid JSON object** with "question" and "answer" (double-quoted keys; valid JSON only; no markdown fences or extra text).

"question":

- Wording refers **only** to the **two temporal** images. Ask for **reasoning-oriented** comparison (how to analyze, what kinds of differences to consider), not yes/no.
- Tell the respondent to **discount** differences from **season, phenology, sun angle, shadows, haze, or global brightness/contrast**, and to focus on **structural change** (new/removed objects, altered built footprint, land-cover conversion, layout), not radiometric or atmospheric appearance alone.
- **No** spatial hints (no compass, edges, "near X", named patches, or "what changed in region X").
- **No** third image, mask, white/black pixels, bands, or file paths.
- High-level process/category terms are OK **without** tying them to a place in the scene.

"answer":

- One string with exactly, in order: <think>...</think> then <answer>...</answer>.
- <think>: Step-by-step reasoning **as if** only Time-1 and Time-2 were seen (appearance, texture, geometry, land cover). **No** mention of (3), mask, ground truth, labels, reference/binary/auxiliary maps, or any hint that a separate change map exists.
- <answer>: Short **what** + **where** in visual/geographic terms; must **agree spatially** with (3) but **still** no (3)-related wording. Put "where" here, not in the question.

In the **entire** "answer" value, the substring "mask" (any case) must not appear.

Figure 13: Prompt issued to the teacher model to distill <think>/<answer> traces from a bi-temporal image pair during QUAKE-CoT construction.

Prompt for Hard Negative Generation

You are an expert in remote sensing change detection answer writing.

****Inputs (in the user message, in order):****

1) Time-1 image (earlier), 2) Time-2 image (later), 3) binary change reference (white=changed, black=unchanged), 4) a reference answer text (Query).

Your task is to generate a new answer whose core factual claim intentionally conflicts with the Query (e.g., what changed, where it changed, or whether meaningful change happened), while preserving the same writing quality profile.

Use the following criteria jointly when imitating writing quality:

- 1) Semantic completeness: keep a comparable coverage of important change details.
- 2) Conciseness and length quality: keep a similar length/granularity; avoid verbosity, repetition, and over-short underspecified text.
- 3) Structural coherence: keep a compatible logical structure (main conclusion, key change factors, and causal/contrast relations).

Important:

- Only change 1-2 core facts (what/where/whether change happened). Keep non-core scene entities and descriptive scaffolding aligned with the Query.
- Keep length close to Query. Target about 0.8x-1.2x of the Query text length.
- Preserve a compatible discourse order with Query.
- Do not mention mask/binary map/ground truth/reference answer/input index.

****Output format (strict, non-negotiable):****

- Output ****nothing**** except one JSON object.
- The object has ****exactly one**** top-level key: ****"response"****.
- The value of ****"response"**** is one JSON string containing exactly:
 - 1) `<think>...</think>`: concise RS-style reasoning.
 - 2) `<answer>...</answer>`: concise what+where statement.
- No markdown fences, no backticks, no extra keys, no extra prose.
- The substring "mask" (any case) must not appear anywhere in the JSON.

Figure 14: Prompt issued to Gemini-3.1-Pro-Thinking-Preview to synthesize hard negatives for the preference hierarchy in §C.

Prompt for Reranker

You are an expert evaluator for remote sensing change detection answers.

Given a reference answer (Query) and a candidate answer (Document), judge whether the Document is truly relevant and faithful to the Query.

Use all of the following criteria jointly:

- 1) Semantic fidelity: preserve the same change semantics and key entities.
- 2) Semantic completeness: do not omit important change details.
- 3) Conciseness and length quality: avoid verbose, repetitive, or overly short responses that lose meaning.
- 4) Structural coherence: keep a compatible logical structure (main conclusion, key change factors, and causal/contrast relations).

Note: all four criteria are equally important. No criterion is optional or secondary.

Semantically correct answers with different wording/order should still be considered relevant.

Superficially similar wording/structure with wrong conclusions or missing key facts should be considered not relevant.

Figure 15: Prompt issued to Qwen3-VL-Reranker to score candidate responses during preference-pair selection.

Prompt for GPT-as-Judge

You are an expert evaluator for remote sensing change detection captions.

Inputs:

1. Ground-truth reference: <think> + <answer>.
2. Model prediction: <think> + <answer>.

Task: decide whether the prediction is semantically consistent with the reference. Prioritize the <answer> fields; use reasoning as supporting context.

Output 1 when the prediction substantially agrees with the reference on:

- Core change type (e.g., new construction, demolition, road/infrastructure, land clearing, renovation, expansion). Synonyms and paraphrases are acceptable.
- Spatial focus: the same general area/region (e.g., upper right, lower left, central). Exact wording need not match; compatible or overlapping locations count.
- Main conclusion: the prediction captures the reference's primary change and does not contradict whether meaningful permanent built-environment / land-use modification occurred.

Minor differences in detail, phrasing, reasoning path, or secondary mentions (e.g., brief extra context in thinking) are acceptable if the <answer> meaning aligns.

Output 0 only for clear mismatches, such as:

- Wrong or incompatible primary change type vs. reference.
- Clearly different scene region when the reference specifies location.
- Missing the reference's main change, or claiming no significant change when the reference describes one.
- Inventing a major change that contradicts the reference.
- Superficial overlap only (shared words but different semantics).

When borderline, lean toward 1 if the answers describe the same event in the same general area; lean toward 0 only when the mismatch is clear.

Output format:

Return ONLY a single integer: 1 (consistent) or 0 (inconsistent). No explanation.

Figure 16: Prompt issued to GPT-5.5 for pairwise quality scoring reported in §4.